

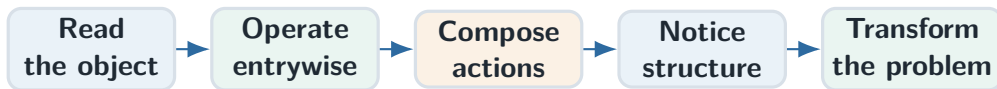
Matrices

Read the shape. Understand the action.

Basic Mathematics ▪ Lecture 1

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 0 & 5 \end{pmatrix}$$

The story of the lecture



A matrix is useful because it keeps structure visible while we calculate.

Three 45-minute arcs

Arc I — Language

shape, entries, rows,
columns

addition and scaling

Arc II — Action

row $\times$ column

composition and order

Arc III — Structure

identity, transpose, diagonal

row operations

1

What matrices are

From a rectangular array to a mathematical object

A matrix can play several roles

What matrices are

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix}$$

data

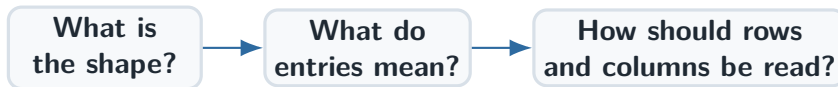
coefficients

relations

transformation

geometric action

The entries alone do not tell the whole story; the interpretation comes from the problem.



Calculation comes after orientation.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

Shape

$$A \in \mathbb{R}^{m \times n}$$

m rows

n columns

Address

$$a_{ij}$$

row i , column j

$$M = \begin{pmatrix} 4 & -2 & 0 \\ 7 & 1 & 5 \end{pmatrix}$$

Shape

$$M \in \mathbb{R}^{2 \times 3}$$

Selected entries

$$m_{11} = 4$$

$$m_{12} = -2$$

$$m_{23} = 5$$

Blue: first row. Green: entry in row 2, column 3.

Row matrix

$$r = \begin{pmatrix} 4 & -2 & 0 \end{pmatrix}$$
$$1 \times 3$$

Column matrix

$$c = \begin{pmatrix} 4 \\ -2 \\ 0 \end{pmatrix}$$
$$3 \times 1$$

Same numbers, different shape, different algebraic role.

row

$$\begin{pmatrix} a & b & c \end{pmatrix}$$
$$1 \times 3$$

column

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$
$$3 \times 1$$

square

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
$$2 \times 2$$

rectangular

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$
$$2 \times 3$$

Shape determines which operations are possible.

- ① name the shape;
- ② locate entries by indices;
- ③ separate rows from columns;
- ④ identify the intended interpretation.

Do not start with

“Which formula should I use?”

Start with

“What object am I holding?”

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Basic matrix operations

Operations that change entries but preserve shape

What remains unchanged?



These operations modify values while keeping every position in place.

Defined

$$(2 \times 3) + (2 \times 3)$$

Every entry has a partner.

Not defined

$$(2 \times 3) + (3 \times 2)$$

Positions no longer match.

Addition is position-by-position matching.

Addition

$$A + B = (a_{ij} + b_{ij})$$

Subtraction

$$A - B = (a_{ij} - b_{ij})$$

Scalar multiplication

$$\lambda A = (\lambda a_{ij})$$

one number acts on every entry

Worked example: addition

$$\begin{array}{|c|} \hline A \\ \hline \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \\ \hline \end{array} + \begin{array}{|c|} \hline B \\ \hline \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix} \\ \hline \end{array} = \begin{array}{|c|} \hline A + B \\ \hline \begin{pmatrix} 7 & 3 \\ -2 & 4 \end{pmatrix} \\ \hline \end{array}$$

$$2 + 5 = 7 \quad -1 + 4 = 3 \quad 0 + (-2) = -2 \quad 3 + 1 = 4$$

$$A - B$$

$$\begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} - \begin{pmatrix} 5 & 4 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & -5 \\ 2 & 2 \end{pmatrix}$$

$$-3A$$

$$-3 \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -6 & 3 \\ 0 & -9 \end{pmatrix}$$

$$\alpha A + \beta B$$

$$2 \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -4 & 4 \end{pmatrix}$$

Interpretation

scale two matrices, then combine
matching positions

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$\lambda(A + B) = \lambda A + \lambda B$$

Reason

Each statement is true entry by entry because ordinary numbers satisfy the same laws.

Entrywise model

This model belongs to addition and scalar multiplication.

Row-column model

Matrix multiplication makes a row interact with a column.

The next operation changes the logic, not only the formula.

- same size is required;
- shape is preserved;
- every entry keeps its address;
- familiar arithmetic laws pass through.

Position is the organizing principle.

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Matrix multiplication

Rows meet columns, and order becomes meaningful

Addition

Matching positions interact.

Multiplication

One row interacts with one column.

A product combines information across positions.

$$(m \times n)(n \times p) = m \times p$$

Inner dimensions

Same length: one row of A meets one column of B .

Outer dimensions

Number of rows and columns in the product.

$$\underbrace{(a_{i1} \quad a_{i2} \quad \cdots \quad a_{in})}_{\text{row } i \text{ of } A} \underbrace{\begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}}_{\text{column } j \text{ of } B} = c_{ij}$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

The whole product is a grid of local interactions

Matrix multiplication

$$C = AB = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{pmatrix}$$

Each location

c_{ij} records row i of A interacting with column j of B .

Global from local

The product is assembled from these row-column answers.

$$A = \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 5 & -1 \end{pmatrix}$$

Shape check

$$(2 \times 2)(2 \times 2) = 2 \times 2$$

Plan

Four output entries require four row-column products.

c_{11}

$$\begin{pmatrix} 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = 1 \cdot 2 + 3 \cdot 5 = 17$$

c_{12}

$$\begin{pmatrix} 1 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = 1 \cdot 0 + 3(-1) = -3$$

The same row produces a full row of the product.

$$c_{21} = (-2) \cdot 2 + 4 \cdot 5 = 16$$

$$c_{22} = (-2) \cdot 0 + 4(-1) = -4$$

Assembled product

$$AB = \begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}$$

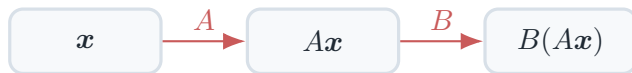
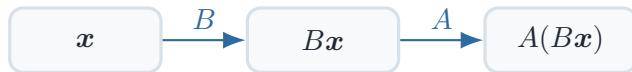
AB

$$\begin{pmatrix} 17 & -3 \\ 16 & -4 \end{pmatrix}$$

BA

$$\begin{pmatrix} 2 & 6 \\ 7 & 11 \end{pmatrix}$$

$AB \neq BA$ in general.



Changing the order changes the process being performed.

Defined

$$A \in \mathbb{R}^{2 \times 3}, \quad B \in \mathbb{R}^{3 \times 1}$$

$$AB \in \mathbb{R}^{2 \times 1}$$

Not defined

$$(3 \times 1)(2 \times 3)$$

The inner dimensions 1 and 2 do not match.

$$A \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^n$$

$$A\mathbf{x} \in \mathbb{R}^m$$

Interpretation

The matrix receives an input vector and produces an output vector.

Shape logic

Input length n must match the number of matrix columns.

$$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n), \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

The output is a linear combination of the columns of A .

Associativity

$$(AB)C = A(BC)$$

Grouping may change.

Order

$$AB \neq BA \quad \text{in general}$$

Order remains fixed.

- ① check inner dimensions;
- ② predict the output shape;
- ③ compute row by column;
- ④ keep the order fixed;
- ⑤ interpret the product as composition.

Each entry is local; the transformation is global.

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Special matrices and visible structure

Patterns are mathematical information, not decoration

$$\begin{pmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Visible immediately

no off-diagonal interaction

Consequence

powers, products and interpretation become transparent

Seeing the pattern is part of solving the problem.

Zero matrix

$$A + 0 = A$$

neutral for addition

Identity matrix

$$I_m A = A, \quad A I_n = A$$

neutral for multiplication

$$A \in \mathbb{R}^{m \times n}$$

Left identity

$$I_m A = A$$

I_m has shape $m \times m$

Right identity

$$A I_n = A$$

I_n has shape $n \times n$

The identity is conceptually one object, but dimensionally many matrices.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$
$$2 \times 3$$

$\longrightarrow$

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$
$$3 \times 2$$

$$\begin{aligned}(A^T)^T &= A \\ (A + B)^T &= A^T + B^T \\ (AB)^T &= B^T A^T\end{aligned}$$

Symmetric matrix

$$A^T = A$$

entries mirror across the main diagonal

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

Visible pattern

Only the main diagonal may contain non-zero entries.

$$D = \text{diag}(3, -1, 2)$$

Second product

$$D^2 = \text{diag}(9, 1, 4)$$

Third product

$$D^3 = \text{diag}(27, -1, 8)$$

Each diagonal position evolves separately.

Zeros

sparse, triangular, diagonal

Repetition

equal rows, columns, blocks

Dependence

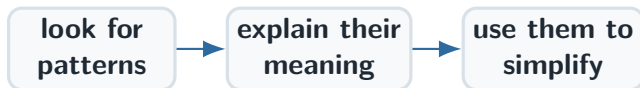
proportional rows or
columns

Symmetry

mirrored entries

Use the pattern

State what becomes simpler
and why.



Noticing structure is mathematical understanding.

5

Row operations and problem solving

Transform the representation to reveal hidden relations

Not the goal

Make the matrix look different.

The real goal

Reveal dependence, simplify structure, prepare a system for solving.

A row operation is a controlled change with a purpose.

Swap

$$R_i \leftrightarrow R_j$$

reorder equations

Scale

$$R_i \leftarrow \lambda R_i, \quad \lambda \neq 0$$

change normalization

Combine

$$R_i \leftarrow R_i + \lambda R_j$$

remove repeated
information

A row operation should tell a story

Row operations and problem solving



The interpretation is the final step: what did the transformation reveal?

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}$$

Observation

The first two entries of R_2 are exactly twice the first two entries of R_1 .

Intention

Remove that repeated part and see what remains.

$$R_2 \leftarrow R_2 - 2R_1$$

$$\underbrace{(2, 4, 3)}_{R_2} - 2 \underbrace{(1, 2, -1)}_{R_1} = (2 - 2, 4 - 4, 3 + 2) = \underbrace{(0, 0, 5)}_{R_2^{\text{new}}}$$

Cancelled

The repeated contribution in columns 1 and 2.

Exposed

A remaining contribution in column 3.

before

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 0 & 1 & 5 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1$$



after

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 5 \\ 0 & 1 & 5 \end{pmatrix}$$

The new row shows exactly what was not already explained by the first row.

1. Arithmetic

subtract two row vectors

2. Transformation

replace R_2 by a new row

3. Interpretation

reveal dependence and residual information

A good solution makes all three layers visible.

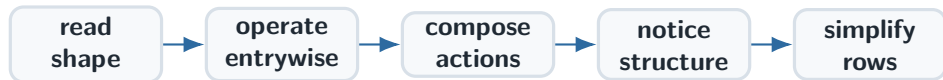
Column form

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \in \mathbb{R}^{3 \times 1}$$

Row form

$$\mathbf{u}^T = \begin{pmatrix} 2 & -1 & 4 \end{pmatrix} \in \mathbb{R}^{1 \times 3}$$

Transpose changes the algebraic role of the same coordinates.



The calculation matters; the interpretation connects the calculation to the problem.

Read the shape.

Then calculate. Then interpret.